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VII SEMESTER

Stream Processing and Analytics (AI372IA)

Phase 1

on

Real-Time Marine Vessel Tracking and Route Analytics

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Introduction

- Over 80% of global trade by volume moves by sea (UNCTAD). Tens of thousands of vessels are at sea at any moment.
- Ships broadcast their position, speed and course through the Automatic Identification System (AIS). This produces millions of messages per day.
- Raw AIS is noisy, incomplete and arrives as a high-velocity stream. Operators struggle to turn it into real-time insight.
- Problems this causes:
  - Collision and grounding risk
  - Port congestion and inaccurate arrival times
  - Illegal fishing, AIS "going dark" and spoofing
  - Route inefficiency and fuel waste
- Our project: a real-time system that ingests live AIS data, tracks vessels, predicts their trajectories, detects anomalies and analyses routes. Users can ask questions in natural language.

Existing System

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- **Current tools:** MarineTraffic, VesselFinder and FleetMon show vessel positions on a map. Coast-guard and port VTS systems use radar plus AIS.
- **Strengths:** live positions, vessel details and basic history.

- **Limitations:**
  - **Rule-based alerts only (geofence, speed threshold)**
  - **Little automatic detection of unusual behaviour**
  - **Poor handling of noisy, missing or spoofed AIS**
  - **No natural-language querying**
  - **Advanced analytics sit behind paid, closed platforms**

Identification of problem

- AIS messages are irregular and have gaps, duplicates and GPS jumps.
- Short-term trajectory prediction is hard: ships turn, slow down and change routes.
- Rule-based anomaly detection gives many false alarms.
- Route patterns (shipping lanes, port-to-port) are not learned automatically.
- No single open pipeline combines real-time streaming, prediction, anomaly detection and analytics.
- Operators cannot ask questions like "Which tankers deviated from their usual route near Singapore today?"

Dataset Description

- Primary sources (open AIS data):

Source	Use
NOAA / MarineCadastre AIS (US coastal waters, daily CSV)	Offline training and EDA
Danish Maritime Authority AIS (open)	Second region, for generalisation testing
AISStream.io live WebSocket feed	Real-time demonstration

Key attributes: MMSI (vessel ID), Timestamp, Latitude and longitude, SOG (speed over ground), COG (course over ground), Heading, Vessel type, Length and width, Draught, Navigational status, Destination and ETA, IMO number

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EDA Results

Dataset Overview

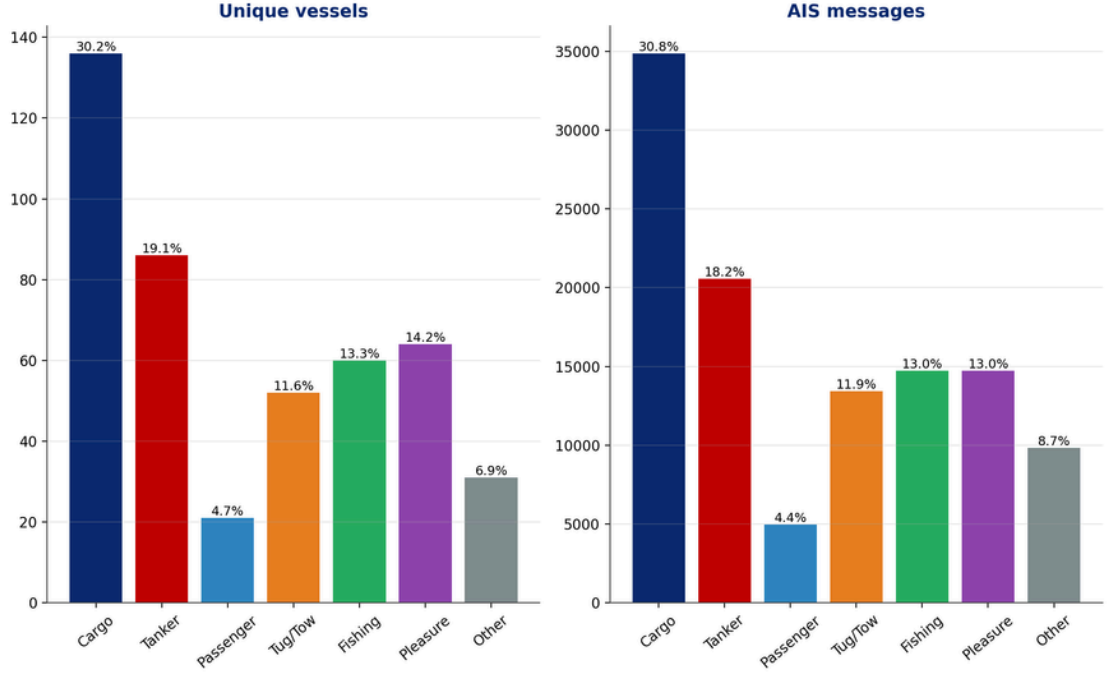
Volume, coverage and vessel population

114,562 Total records	**450** Unique vessels (MMSI)	**74.7 h** Time span
1.19% Duplicate messages	**36** Invalid coordinates	**7** Vessel classes

Metric	Value
Latitude range	26.90 to 30.53
Longitude range	-97.46 to -82.54
Start	2024-01-15 00:02:10
End	2024-01-18 02:44:44
Median msgs / vessel	225
Median reporting gap	94 s

Vessel Type Distribution

Fleet is cargo / tanker heavy; small craft (tug, fishing, pleasure) send proportionally more messages

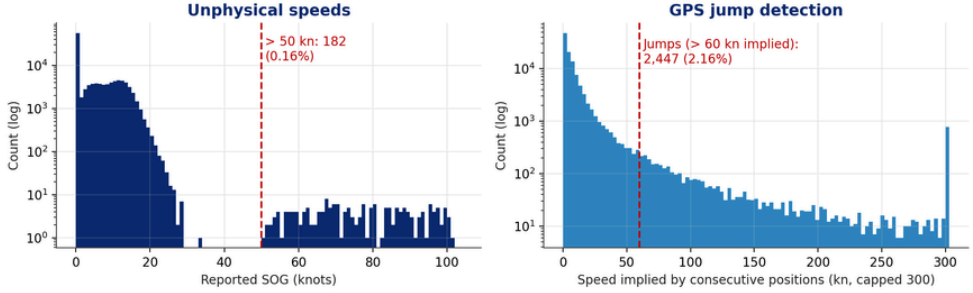


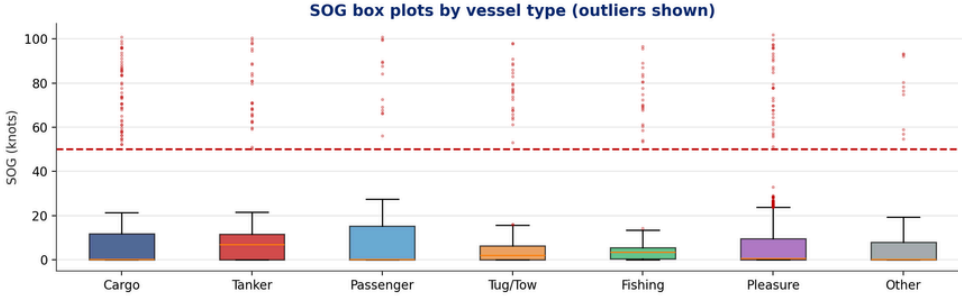
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EDA Results

Outliers and Noise Profiling

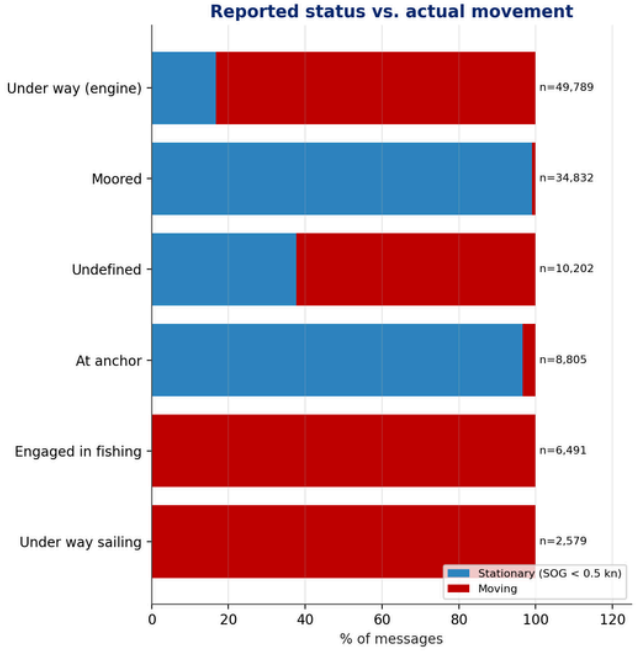
Invalid coordinates found: 36 rows (lat 91 / lon 181 sentinel values)

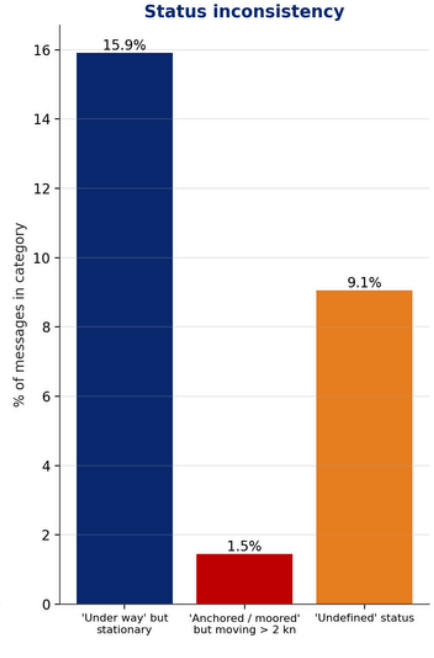




Navigational Status vs. Actual Movement

Self-reported status is unreliable, so anomaly features should use kinematics, not the status field





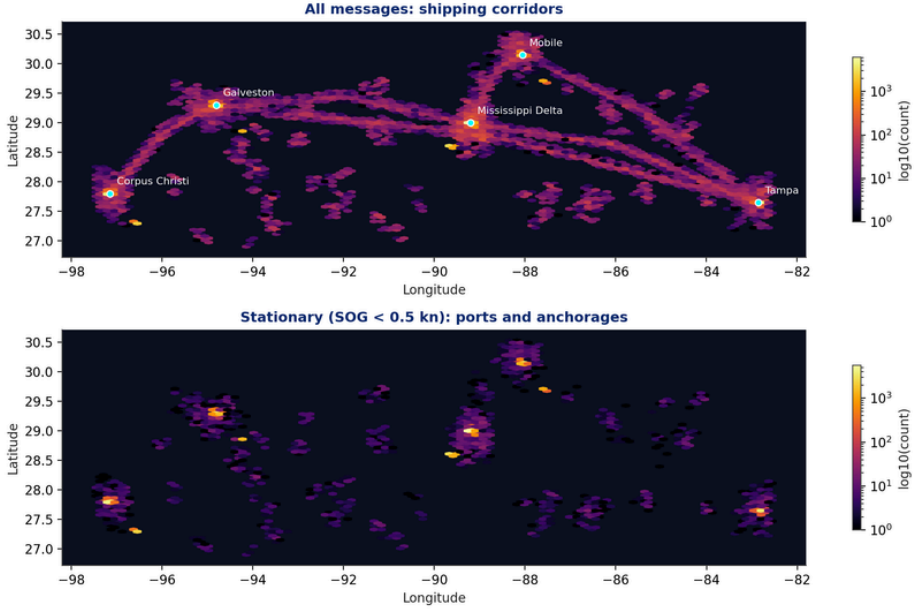
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EDA Results

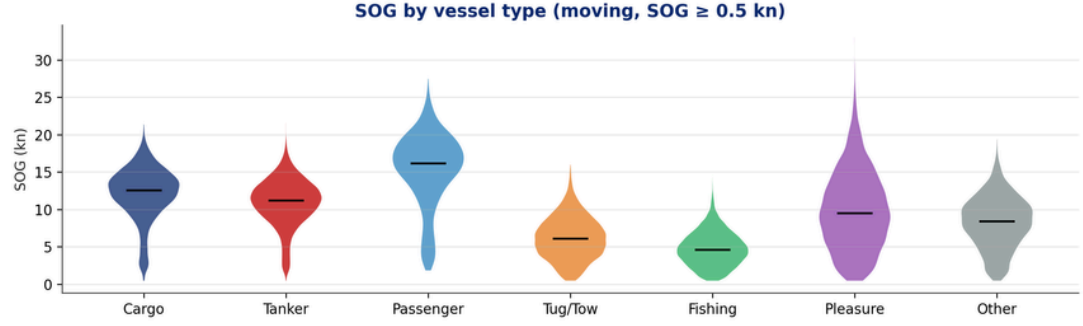
Spatial Density of Traffic

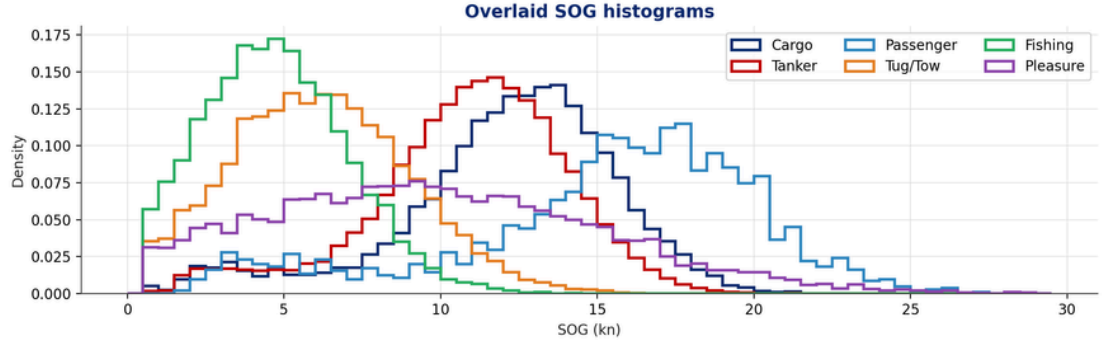
Clear lanes between ports; tight hotspots at ports and anchorages justify DBSCAN route clustering



Speed Over Ground by Vessel Type

Cargo and tankers cruise steadily; tugs and fishing boats are slow and variable, so type is a strong model feature



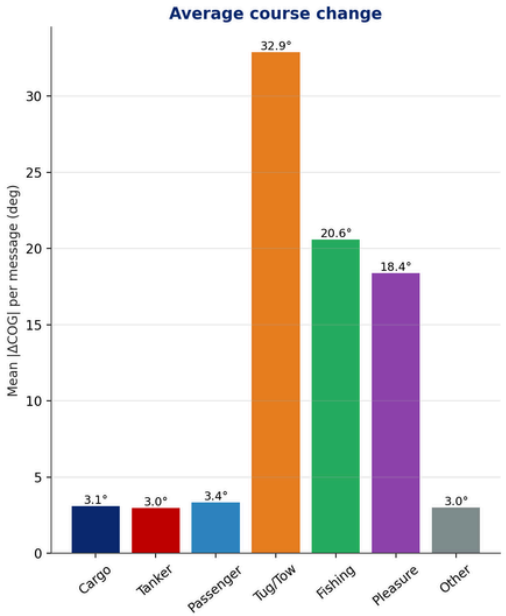


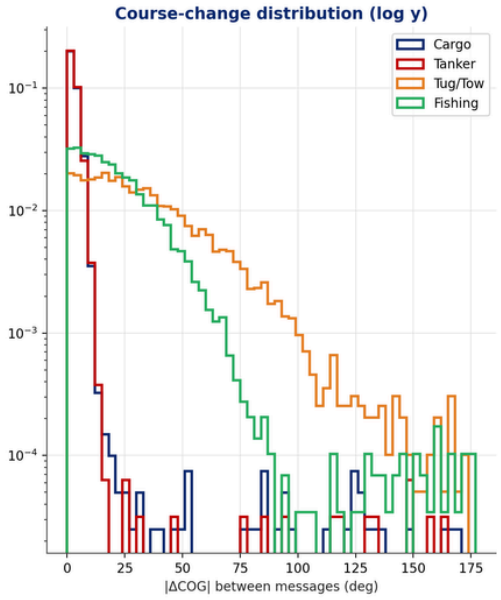
EDA Results

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Manoeuvring Behaviour by Vessel Type

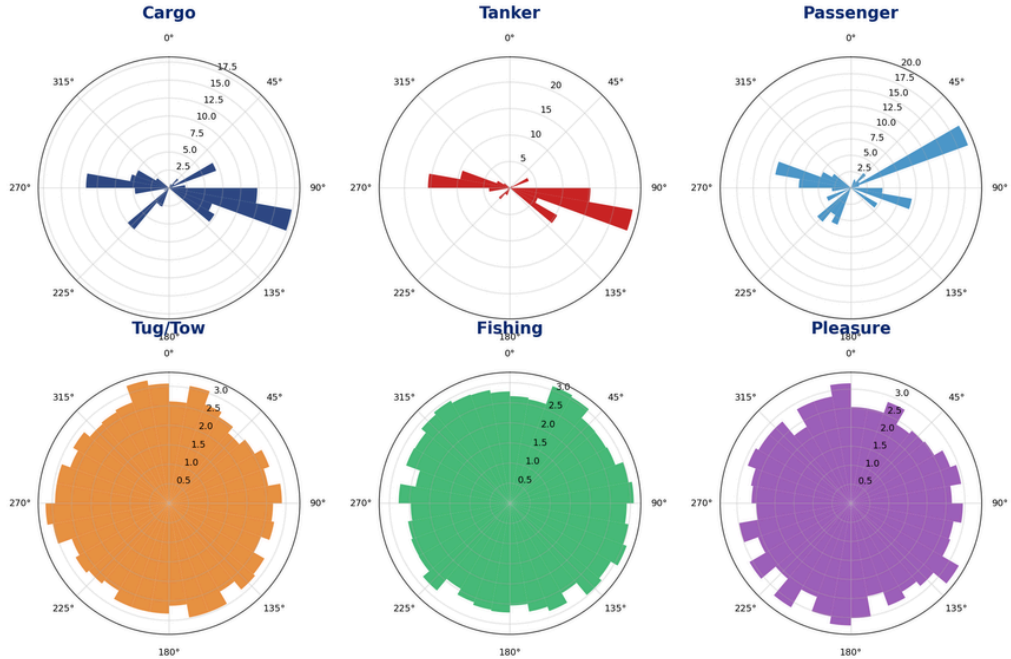
Large turn rates are normal for tugs and fishing boats, so a single global rule would cause false alarms





Course Over Ground Distribution

Lane traffic forms sharp directional peaks; tugs / fishing / pleasure craft spread over all headings (moving vessels only)



Literature Review

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#	Paper	Summary	Gaps
1	Pallotta, Vespe, Bryan, "Vessel Pattern Knowledge Discovery from AIS Data," *Entropy*, 2013	Unsupervised extraction of traffic routes (TREAD) with anomaly detection and route prediction	Uses statistical and clustering rules, so it cannot capture complex temporal patterns
2	Nguyen et al., "GeoTrackNet," *IEEE T-ITS*, 2022	Probabilistic neural network (VRNN) plus a-contrario detection for anomalies in AIS tracks	Detection only, offline, and tuned for one region
3	Nguyen and Fablet, "TrAISformer: A Generative Transformer for AIS Trajectory Prediction," arXiv 2021	Transformer with discretised "four-hot" position encoding; beats LSTM baselines	Heavy compute, no live streaming, evaluated on limited regions
4	Forti et al., "Prediction of Vessel Trajectories from AIS Data via Seq2Seq RNNs," *ICASSP*, 2020	Encoder-decoder RNN for vessel trajectory prediction	Short horizons, no interaction between vessels
5	Capobianco et al., "Deep Learning Methods for Vessel Trajectory Prediction Based on RNNs," *IEEE Trans. AES*, 2021	Compares RNN models and gives uncertainty estimates for prediction	Does not use environment or port context

Literature Review

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#	Paper	Summary	Gaps
6	Murray and Perera, "An AIS-based deep learning framework for regional ship behavior prediction," *Reliability Eng. & System Safety*, 2021	Clusters trajectories, then uses a regional BiLSTM for behaviour prediction	Region-specific, with a fixed clustering step
7	Sheng and Yin, "Extracting Shipping Route Patterns by Trajectory Clustering Model Based on AIS Data," *ISPRS IJGI*, 2018	Clustering model that discovers route patterns	Offline batch method, sensitive to parameters
8	Kontopoulos et al., "A distributed framework for extracting maritime traffic patterns," *IJGIS*, 2021	Scalable (distributed) pipeline for traffic-pattern extraction	Focus on patterns, not prediction or live use
9	Riveiro, Pallotta, Vespe, "Maritime anomaly detection: A review," *WIREs Data Mining & Knowledge Discovery*, 2018	Surveys anomaly-detection methods, such as statistical, clustering and rule-based	A review only, with no common benchmark or real-time solution
10	Wolsing et al., "Anomaly Detection in Maritime AIS Tracks: A Review of Recent Approaches," *J. Marine Sci. & Eng.*, 2022	Reviews recent ML approaches and the lack of labelled data	Points out that ground-truth anomalies are scarce

Literature Review

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#	Paper	Summary	Gaps
11	Ester et al., "DBSCAN," *KDD*, 1996	Density-based clustering that finds arbitrary-shaped clusters and noise	Fixed parameters, no time dimension
12	Hochreiter and Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997	Gated recurrent unit that learns long-range dependencies	Slow sequential training, weaker on long sequences than Transformers
13	Vaswani et al., "Attention Is All You Need," *NeurIPS*, 2017	Self-attention architecture that replaced recurrence	Quadratic cost in sequence length, needs adaptation for geospatial data
14	Kalman, "A New Approach to Linear Filtering and Prediction Problems," *J. Basic Eng.*, 1960	Optimal recursive state estimation, widely used for tracking	Assumes linear motion and Gaussian noise
15	Chen and Guestrin, "XGBoost," *KDD*, 2016	Scalable gradient-boosted trees, strong on tabular data	Does not model sequences directly, needs engineered features

Literature Review

#	Paper	Summary	Gaps
16	Redmon et al., "You Only Look Once," *CVPR*, 2016	Real-time single-stage object detector	Struggles with small objects such as distant ships
17	Shao et al., "SeaShips: A Large-Scale Precisely Annotated Dataset for Ship Detection," *IEEE Trans. Multimedia*, 2018	Benchmark of annotated ship images from coastal surveillance cameras	Mostly nearshore, with limited weather and night variety
18	Bewley et al., "Simple Online and Realtime Tracking (SORT)," *ICIP*, 2016	Kalman filter plus Hungarian assignment for tracking	Many identity switches under occlusion
19	Wojke et al., "Simple Online and Realtime Tracking with a Deep Association Metric (DeepSORT)," *ICIP*, 2017	Adds appearance embeddings to SORT, which cuts identity switches	Needs a good re-ID model, with extra compute
20	Zhang et al., "ByteTrack: Multi-Object Tracking by Associating Every Detection Box," *ECCV*, 2022	Associates low-confidence detections too, giving state-of-the-art accuracy	Built for video, so it needs adapting to maritime scenes

Literature Review

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#	Paper	Summary	Gaps
21	Carion et al., "End-to-End Object Detection with Transformers (DETR)," *ECCV*, 2020	Transformer detector with no anchors or NMS	Slow convergence, weaker on small objects
22	Liu et al., "Swin Transformer," *ICCV*, 2021	Hierarchical vision Transformer with shifted windows, a strong detection backbone	High memory use, needs large data
23	Kipf and Welling, "Semi-Supervised Classification with Graph Convolutional Networks," *ICLR*, 2017	Learns on graph-structured data	Static graphs, not designed for streaming data
24	Wu et al., "Graph WaveNet for Deep Spatial-Temporal Graph Modeling," *IJCAI*, 2019	Spatio-temporal graph model for traffic forecasting	Built for road networks, and applying it to ports or sea lanes is open
25	Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *NeurIPS*, 2020	Combines retrieval with a generator to ground answers in data	Not yet applied to live maritime data and analytics queries

Summary of Literature Review

What exists:

- Clustering for route discovery (DBSCAN, TREAD)
- RNN/LSTM, and recently Transformer, models for trajectory prediction
- VAE and statistical methods for anomaly detection
- YOLO with SORT/DeepSORT/ByteTrack for visual vessel tracking
- Surveys that cover the whole anomaly-detection field

Research gaps:

- Most work is offline and batch. Real-time streaming is rare.
- Prediction, anomaly detection and route analytics are studied separately and not combined.
- Models are trained on one region and don't generalise.
- Labelled anomaly data is scarce.
- Transformers have shown promise (TrAISformer), but deployed live systems are few.
- No natural-language interface, although RAG is proven in other fields.

Problem Definition

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Design and build a real-time system that ingests streaming AIS data and tracks vessels. It should predict their future trajectories and arrival times, detect abnormal behaviour, and discover route patterns. The system should present the results on a live dashboard and let users query them in natural language.

Objectives

- To build a real-time pipeline that ingests live AIS data, cleans it, and reconstructs vessel tracks.
- To predict short-term vessel trajectories (30–60 min) with LSTM and Transformer models, and compare them.
- To detect anomalies (route deviation, sudden stops, AIS gaps) and discover route patterns by clustering.
- To provide a live dashboard with a natural-language query assistant (RAG and LLM), and evaluate accuracy and latency.

Methodology

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1. Data Acquisition (AIS Data Sources)

AISStream.io
(Live WebSocket)
Real-time AIS data

NOAA / MarineCadastre
(Daily CSV)
Offline training & EDA

Danish Maritime Authority
(Open AIS)
Generalisation testing

Key Attributes
- MMSI (vessel ID)
- Timestamp
- Latitude, Longitude
- SOG (speed), COG (course)
- Heading, Vessel type
- Destination, ETA
- Length, Width, Draught
- Navigational status
- IMO number

2. Kafka Ingestion (Producer & Topics)

Python Producer
(Reads live WebSocket / CSV and publishes AIS events)

Apache Kafka
(Broker Cluster)

Topic 1: ais_raw
(Raw AIS events)

```
{
 "mmsi": 123456789,
 "timestamp":"...",
 "lat":1.29,
 "lon": 103.85,
 "sog": 12.4,
 "cog": 87.6,
 "type": "tanker"
} ...
```

Topic 2: ais_alerts
(Detected anomalies)

```
{
 "mmsi": 123456789,
 "timestamp":"...",
 "alert_type": "ROUTE_
  DEVIATION",
 "severity":"HIGH",
 "details":"..."
}
```

Partitioning (by MMSI)

Partition 0	Partition 1	Partition 2	Partition 3
(MMSI A,D,G)	(MMSI B,E,H)	(MMSI C,F,I)	(...)

3. Kafka Consumer Group (Multiple Consumers)

Consumer Group: ais-processing-group

Consumer 1	Consumer 2	Consumer 3
(Process Partition 0)	(Process Partition 1)	(Process Partition 2)

Load balancing & parallel processing
(Automatic rebalance on failure)

Fault Tolerance & Recovery
- Kafka replication for broker fault tolerance
- Consumer group rebalances on consumer failure
- Offsets stored in Kafka (resume from last processed position)
- At-least-once processing guarantee

4. Stream Processing (Spark Structured Streaming)

Data Cleaning & Preprocessing
- Remove duplicates
- Handle missing values
- Validate lat/lon (valid range)
- Check impossible speeds
- Detect GPS jumps
- Normalize timestamps
- Identify AIS gaps
- Sort by timestamp

Window-based Processing
(5-minute sliding window)

Window 1 (0-5 min)	Window 2 (1-6 min)	Window 3 (2-7 min)

Slide every 1 min

5. Real-Time Aggregation & Stream Analytics

Aggregation / Summarization
- Number of active vessels (count unique MMSI)
- Average vessel speed (per region)
- Vessel count by type (tanker, cargo, ...)
- Vessel density (per region/grid)
- Number of anomalies (per window)
- Top ports (arrivals/departures)

Example Output (5-min window)

Active Vessels	1,245
Avg Speed	12.4 kn
Tankers	342
Cargo	521
Anomaly Count	17

6. Vessel Track Reconstruction & Feature Engineering

Tracks per MMSI
(maintain recent history)



Derived Features
- Distance travelled
- Average speed
- Acceleration / deceleration
- Change in course
- Turning rate
- Time since last message
- Route deviation
- AIS gap duration

7. Trajectory Prediction (30–60 min ahead)

Input:
Past 20-30 AIS points (lat, lon, SOG, COG, heading, time, ...)

LSTM Model

Transformer Model

Predicted Trajectory
(future positions)

8. ETA Prediction

Predicted trajectory + vessel speed

Destination (from AIS)

Distance & time calculation

Estimated ETA
Compare with AIS ETA (when available)

9. Anomaly Detection

Rule-based (Real-time)
- Speed > threshold
- Sudden stop
- Large course change
- Route deviation
- AIS gap / going dark
- Entering restricted area

ML-based (Isolation Forest)

Features:
- Speed
- Acceleration
- Course change
- Route deviation
- AIS gap duration
- Context features

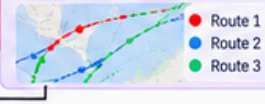
Anomaly Score & Alert

10. Route Pattern Discovery

Historical AIS trajectories

DBSCAN Clustering

Shipping route clusters (common routes / lanes)



11. Storage Layer

PostgreSQL / PostGIS
(Real-time results, spatial queries)
- Current vessel state
- Predicted trajectories
- Anomalies
- Aggregated statistics

Historical Database
(For model training and analysis)
- Cleaned AIS data
- Route clusters
- Model training datasets

12. Real-Time Visualization Dashboard



13. Natural Language Query Interface

"Which tankers deviated from their usual route near Singapore today?"

LLM (with RAG)

Query Generation (filter + analytics)

Results from Database / Stream State

Natural Language Answer

"3 tankers deviated from their usual route near Singapore today. Here are their details ..."

End-to-End System Flow

AIS Source (Live/Offline) → Kafka Producer → Kafka Topics (ais_raw, ais_alerts) → Partitioning (by MMSI) → Consumer Group → Stream Processing (Spark) → Window + Aggregation → ML Analytics (Prediction, Anomaly, Clustering) → Storage (PostgreSQL/PostGIS) → Dashboard & Alerts → Natural Language Query

Methodology

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Algorithm	Pros	Cons	Why not as suitable for our AIS project?
DBSCAN	Finds curved routes; detects noise; no need to specify K	Sensitive to parameters	Chosen: suitable for discovering irregular shipping lanes and identifying isolated GPS points
K-Means	Fast and simple	Needs K; sensitive to outliers	Shipping lanes are curved, so clusters around centres may split routes incorrectly
OPTICS	Handles varying densities; finds irregular clusters	More complex	Good alternative, but adds complexity to our initial implementation
HDBSCAN	Handles varying densities and noise	More complex to tune and explain	Useful if dense port routes and sparse ocean routes are difficult for DBSCAN to separate
Hierarchical	Shows relationships between clusters	Can be slow and memory-intensive	Large AIS datasets may make repeated clustering expensive
GMM	Gives probability of cluster membership	Needs number of components; assumes distribution shapes	Curved shipping lanes may not fit standard Gaussian clusters well
Mean Shift	Finds dense regions without specifying K	Bandwidth-sensitive; can be slow	Large volumes of AIS points may increase processing time
Spectral Clustering	Finds complex cluster shapes	Computationally expensive	Repeatedly clustering large AIS datasets may be impractical for near-real-time analytics
BIRCH	Memory-efficient for large datasets	May lose cluster details	May not preserve irregular shipping-lane shapes or identify noise effectively

Methodology

Clustering Algorithn Justification-

We choose DBSCAN as a practical starting point because it discovers irregular shipping lanes without requiring a predefined number of routes and can identify noise. OPTICS or HDBSCAN would be alternatives if route densities vary significantly.

Conclusion

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The proposed system addresses critical challenges in modern maritime monitoring by establishing a unified, open, end-to-end processing pipeline capable of handling noisy, high-velocity AIS data streams in real time. By combining Apache Kafka for event-driven ingestion and Spark Structured Streaming for real-time cleaning and windowed feature engineering, the architecture turns raw positional broadcasts into reliable vessel tracks. Advanced machine learning modules including LSTM and Transformer architectures for 30–60 minute trajectory and ETA prediction, Isolation Forest for ML-driven anomaly detection, and DBSCAN for automatic route pattern discovery overcome the limitations of traditional, rule-based tracking systems.

Ultimately, this system transforms raw geospatial data into actionable maritime situational awareness, significantly improving collision prevention, port congestion management, and anomaly monitoring such as AIS spoofing or route deviations. By serving these analytics through a live interactive dashboard coupled with a Retrieval-Augmented Generation (RAG) LLM assistant, operators can query complex streaming data in natural language. This seamless bridge between real-time data engineering, deep learning analytics, and conversational AI provides maritime authorities with an accessible, highly scalable tool to optimize fleet efficiency and strengthen maritime security.

References

[1] N. Pallotta, M. Vespe, and K. Bryan, "Vessel Pattern Knowledge Discovery from AIS Data," Entropy, 2013.

[2] D. Nguyen et al., "GeoTrackNet," IEEE T-ITS, 2022.

[3] D. Nguyen and R. Fablet, "TrAISformer: A Generative Transformer for AIS Trajectory Prediction," arXiv, 2021.

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[7] P. Sheng and J. Yin, "Extracting Shipping Route Patterns by Trajectory Clustering Model Based on AIS Data," ISPRS IJGI, 2018.

[8] B. Wolsing et al., "Anomaly Detection in Maritime AIS Tracks: A Review of Recent Approaches," J. Marine Sci. & Eng., 2022.

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Thank You